Exercise 1: Zip

Suppose we have two lists, x and y that give the x and y coordinates of a set of points. Create a list with the coordinates (x,y) as a tuple. Hint: Find out about the zip function. You have decided that actually you need the two separate lists, but unfortunately, you have thrown them away. How can we use zip to unzip the list of tuples to get two lists again?

Exercise2: Distances

Suppose we have two vectors, x and y, stored as tuples with n elements. Implement functions that compute the l1 and l2 distances between x and y. Note that n is not explicitly given.

Exercise 3: Histogram

Write a function that takes a list, and returns a dictionary with keys the elements of the list and as value the number of occurrences of that element in the list.

Exercise 4: Reverse look-up

Dictionaries are made to look up values by keys. Suppose however, we want to find the key that is associated with some value. Write a function that takes a dictionary and a value, and returns the key associated with this value. What challenges do you face? How would you deal with those challenges?

Exercise 5: Vector functions

Let's implement some vector functions. There are two types of vectors, normal or dense vectors, which we can represent using lists. For sparse vectors, where many of the elements are zero, this is inefficient. Instead, we use a dictionary with keys the indices of non-zero values, and then the value corresponding to the key is the value of the vector at that index. Hence, the vector [1; 2; 4] can be stored as a list: [1, 2, 4] or as a dictionary {0:1, 1: 2, 2: 4}

(a) Write a function that adds two (dense) vectors
(b) Write a function that multiplies (i.e. inner product) two (dense) vectors
(c) Write a function that adds two sparse vectors
(d) Write a function that multiplies two sparse vectors
(e) Write a function that adds a sparse vector and a dense vector
(f) Write a function that multiplies a sparse vector and a dense vector